

How to Prove It

2.3 More operations on sets

Example 2.3.3 asks us to analyse the logical forms of several statements involving powersets.

§ **Example 2.3.3**

Parameters

+ $A \in \text{Set}$

+ $B \in \text{Set}$

Observations

1. $x \in \mathcal{P} A$

$$\equiv (\forall z \mid z \in x : z \in A)$$

2. $\mathcal{P} A \subseteq \mathcal{P} B$

$$\equiv (\forall X \mid (\forall z \mid z \in X : z \in A) : (\forall z \mid z \in X : z \in B))$$

3 $B \in \{\mathcal{P} A \mid A \in \mathcal{F}\}$

$$\equiv (\exists A \mid (\forall z \mid z \in A : z \in \mathcal{F}) : (\forall z \mid z \in B \equiv (\forall y \mid y \in z : y \in A)))$$

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